



MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND INNOVATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

COMPUTER HARDWARE REPAIRS AND MAINTENANCE MANUAL

COMPUTER HARDWARE SERVICING

DETAILED SYLLABUS

UNIT	NAME OF THE TOPIC	HOURS
1	<u>MOTHERBOARD COMPONENTS AND MEMORY STORAGE DEVICES</u> Introduction: Hardware, Software and Firmware. Mother board, IO and memory expansion slots, SMPS, Drives, front panel and rear panel connectors. Processors: Architecture and block diagram of multicore Processor, Features of new processor (Definition only)-chipsets (Concepts only). Bus Standards: Overview and features of PCI, AGP, PCMCIA Primary Memory: Introduction-Main Memory, Cache memory – DDR2, DDR3, DDR4 and Direct RDRAM. Secondary Storage: Hard Disk – Construction – Working Principle Specification of IDE, Ultra ATA, Serial ATA; HDD Partition - Formatting. Removable Storage: CD-R, CD-RW, DVD –ROM and DVD –RW: construction and reading & writing operations; Blue-ray – Introduction –Disc Parameters.	
2	<u>I/O DEVICES AND INTERFACE</u> Keyboard: Signals – operation of membrane and mechanical keyboards–troubleshooting; wireless Keyboard. Mouse: types, connectors, operation of Optical mouse and Troubleshooting. Printers: Introduction – Types of printers- Dot Matrix, Inkjet Laser, MFP (Multi-Function Printer) and Thermal printer – Operation, Construction and Features-Troubleshooting.	

	I/O Ports: Serial, Parallel, USB, Game Port and HDMI. Displays: Principles of LED, LCD and TFT Displays. Graphic Cards: VGA and SVGA card. Modem: Working principle. Power Supply: Servo Stabilizers, online and offline UPS - working principles; SMPS: Principles of Operation and block diagram of ATX Power supply, Connector Specifications.	
3	<u>MAINTENANCE AND TROUBLE SHOOTING OF DESKTOP AND LAPTOPS</u> Bios-setup: Standard CMOS setup, Advanced BIOS setup, Power management, advanced chipset features, PC Bios communication – upgrading BIOS, Flash BIOS -setup. POST: Definition – IPL hardware – POST Test sequence – beep codes. Diagnostic Software and Viruses: Computer Viruses – Precautions –Anti-virus Software – identifying the signature of viruses – Firewalls and latest diagnostic software. Laptop components: Adapter – types, Battery – types, Laptop Mother Board - block diagram, Laptop Keyboard and Touchpad. Installation and Troubleshooting: Formatting, Partitioning and Installation of OS – Trouble Shooting Laptop Hardware problems - Preventive maintenance techniques for laptops.	

Rationale:

The course aims at making the students familiar with various parts of computers and how to assemble them and the different types of peripherals desired. In addition, the course will provide the students with necessary knowledge and skills in computer software installation and maintenance and to make them diagnose the software faults. This will provide students with hands on training regarding the design, troubleshooting, modeling and evaluation of computers. Students are going to experiment in a real test-based computing environment.

Objectives

- Knowledge of Computer Hardware Devices
- Identify computer hardware Issues/Problems
- Study of basic computing configuration commands
- Know Basic computer troubleshooting tips
- Configure Bios set up options.
- Identify the problems in Computer systems, software installation and rectification
- Assembling and disassembling of Laptop to identify the parts and to install OS and configure it.
- Knowledge of Operating System and device Drivers
- Basic knowledge of safeguarding hardware
- Assembling and disassembling of Desktops and Laptops to identify the parts and to install OS and configure it.

Experiment NO: -1

OBJECTIVE: - Identification of Hardware elements.

HARDWARE & SYSTEM SOFTWARE REQUIRED: -Personal computer which contains different types of ports,

SOFTWARE REQUIRED: - No software needed

Computer hardware is the collection of physical components that constitute a computer system. Examples of hardware which is used in computer,

1. Monitor
2. Keyboard

3. Computer Data Storage
4. Hard Disk Drive (HDD)
5. Graphic Card
6. Sound Card
7. Memory (RAM)
8. Motherboard

Below is a list of **external hardware and internal hardware** and the categories of hardware that may be found with a computer

External hardware examples

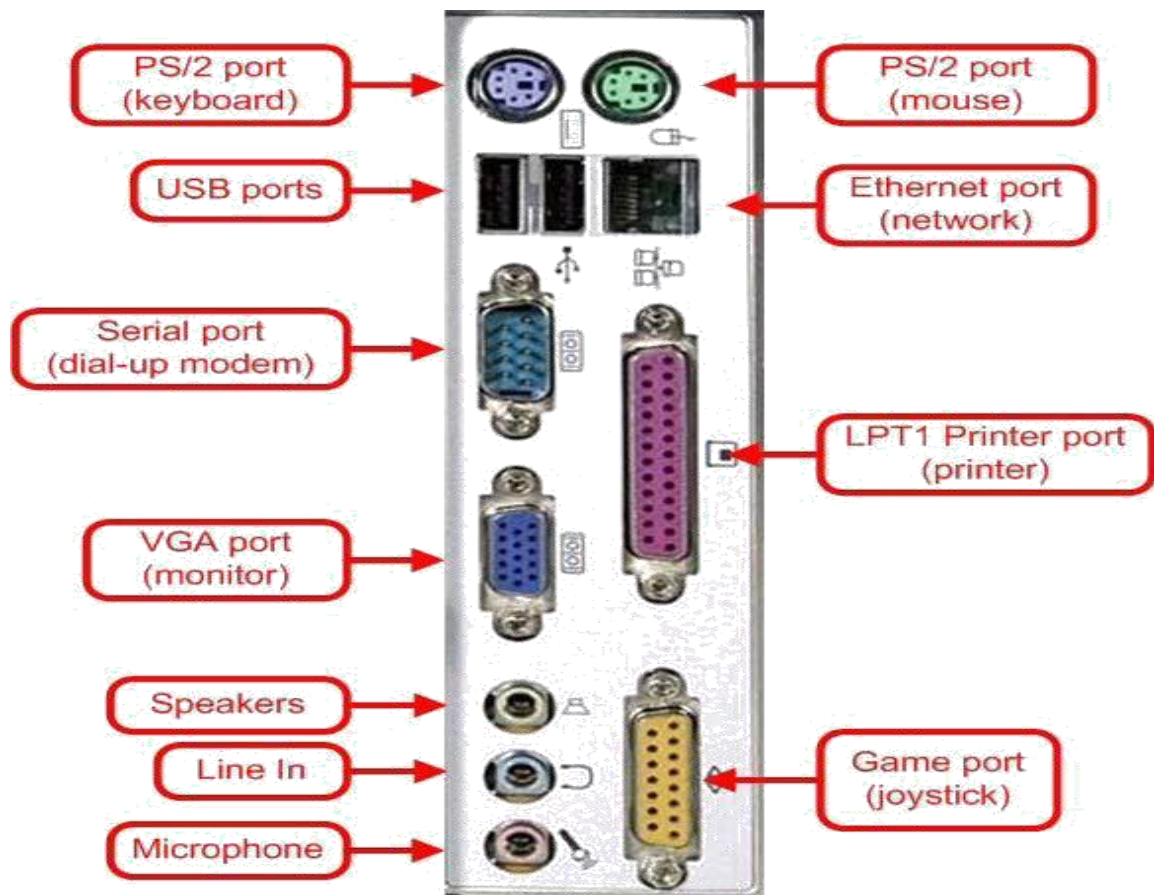
- Flat-panel, Monitor, and LCD
- Keyboard
- Microphone
- Mouse
- Printer
- Projector
- Scanner
- Speakers
- USB thumb drive

Internal hardware examples

- Processor (CPU)
- Drive (e.g. Blu-ray, CD-ROM, DVD, floppy drive, hard drive, and SSD)
- Fan (heat sink)
- Modem
- Motherboard
- Network card
- RAM
- Sound card
- Video card

THEORY: - To learn about different ports and how to connect devices to them

This diagram shows different ports available on the back panel of the PC



Parallel port (LPT parallel port):

As shown in the diagram parallel port with 25-pins can be used to connect a Parallel port printer. Previously dot matrix, ink jet, bubble jet printers etc. were Connected to parallel port. Nowadays-parallel port is used to connect Dot-Matrix Printers.

Serial port:

As shown in the diagram serial ports with 9-pins protruding outwards can be used to connect modem but it can also be used for connecting mouse, provided serial port mouse is available.

VGA Port:

VGA port which has 15-pins is used to connect a monitor.

PS/2 Port:

Two 6-pin PS/2 ports are there, one is violet to which keyboard is connected and other is Light green to which mouse is connected

USB Port:

Connecting a USB device to a computer is simple — you find the USB connector on the back of your machine and plug the USB connector into it. USB ports are used to connect to Inkjet printers, Web Cams, Scanners etc.

Ethernet Port:

Ethernet port is used to connect a computer on network through RJ-45 connector

Game Port:

Game Port is used to connect joystick, which is usually used in video games **Three more ports** are available for multimedia connections. Green port is used connect speakers, blue port is used to connect headphones and light Orange is used to connect microphone.

EXPERIMENT NO: -2

OBJECTIVE: - To Assembling a computer.

HARDWARE & SYSTEM SOFTWARE REQUIRED: -Personal computer which contains different types of ports,

SOFTWARE REQUIRED: -No software required.

THEORY: - To enter into CMOS setup continue pressing Delete (Del) button while system is booting.

This will differ depending on models of the computer. E.g., Dell, Compaq, Acer, Hp and Lenovo has different Bios setup options.

Quick boot: Set this option to enabled

1st Boot Device: This option sets the type of device for the first boot drives that the AMIBIOS attempts to boot from.

2nd Boot Device: This option sets the type of device for the second boot drives that the AMIBIOS attempts to boot from.

3rd Boot Device: This option sets the type of device for the third boot drives that the AMIBIOS attempts to boot from

OBSERVATIONS: --The various features available in CMOS have been to be identified.

EXPERIMENT NO: - 3

OBJECTIVE: - Formatting disk drives.

HARDWARE & SYSTEM SOFTWARE REQUIRED: -Personal computer which contains different types of ports,

SOFTWARE REQUIRED: -windows xp,/windows 2000, windows 7, 8 ,10 and 11 software.

THEORY: -

Hard Disk Partitions

This procedure explains how to setup a new hard disk.

Before a new hard disk can be used it needs to be setup. This involves partitioning and formatting the hard disk. Windows boot disk contains the required Software to perform this procedure. FDISK.EXE and

FORMAT.COM are the files required in your bootable Hard disk or USB drive.

Start the partition and format procedure by booting your PC using a Windows boot disk. Make sure you set the BIOS so that the boot sequence is set to detect the floppy disk first.

If your system has no problems booting you will be presented with a Windows boot disk menu. This gives you the option to start the system with or without CD-ROM support.

At this stage you do not need the CD-ROM support, so choose the option to boot without CD-ROM support. You should end up in the MS DOS prompt A: (A drive).

HDD Partition

Disk partitioning or disk slicing is the technique for creating one or more regions on a hard disk or other secondary storage, so that an operating system can manage information in each region separately. Each partition then appears in the operating system as a distinct "logical" disk that uses part of the actual disk.

Advantages

- Hard drive to manage data in category wise.
- Portioning a hard drive is advantageous to the virus prevention and data security.
- Use the hard disk efficiently.
- Improving the efficiency of the system.
- Assign different permissions for different users.
- Helpful to install multiple operating systems.

Formatting

Disk formatting is the configuring process of a data storage media such as a hard disk drive, floppy disk or flash drive for initial usage. Any existing files on the drive would be erased with disk formatting. Disk formatting is usually done before initial installation or before installation of a new operating

system. Disk formatting is also done if there is a requirement for additional storage in the computer. There are two types of disk formatting,

- Low-Level formatting
- High – Level formatting

Low-Level formatting

Low-level formatting is the process of marking out cylinders and tracks for a blank hard disk, and then dividing tracks into multiple sectors. This process is often called the “*real*” formatting since it creates physical format which defines where the data is saved. If users perform low-level formatting when data have been installed, all existing files will be erased, and it is almost impossible to recover them.

High - Level formatting

After a low-level formatting has been completed, users need to make high-level formatting which makes it possible to save data and should be done on a partition. It is the process of writing a file system, cluster size, partition label, and so on for a newly created partition or volume. This process does no harm to hard disk in general situations, so we suggest taking such a format to fix a logically damaged partition or device, for example, Windows asks to format a SD card.

EXPERIMENT NO: -4

OBJECTIVE: - Identification of symptoms for failure:

- Beep codes and error codes.
- Different symptoms of motherboard.
- Beep, error and symptoms failure of FDDI.
- Keyboard interface.
- Beep, error and symptoms of failure of keyboard.

HARDWARE & SYSTEM SOFTWARE REQUIRED: -Personal computer which contains different types of ports,

SOFTWARE REQUIRED: -No software required.

THEORY: - A sound card or audio board, which allows computers to output audio signals Through speakers and or headphones.

EXPERIMENT NO: -5

OBJECTIVE: - Understanding control panel settings.

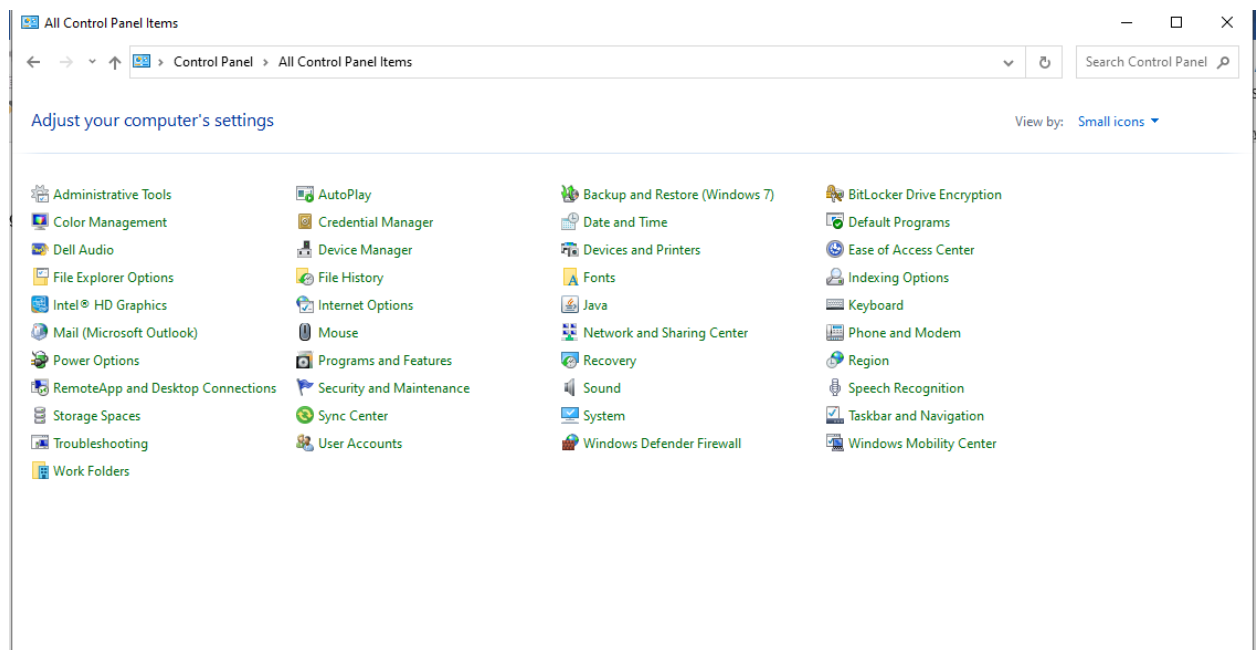
HARDWARE & SYSTEM SOFTWARE REQUIRED: -Personal computer which contains different types of ports,

SOFTWARE REQUIRED: -No software required.

THEORY: - To access a control panel go through the following steps below

Go to start menu —————> select control panel

Figure below shows the options available in control panel.



Accessibility Options: You find keyboard, sound, display, mouse, and general settings available. Using this you can change the behavior of keyboard, mouse, sound, display and mouse.

Add New Hardware:

With this feature you can add new hardware to your computer.

Add / Remove Software:

With this feature you can add new software and remove existing ones.

Network:

With this feature you can add and remove Network Interface Card settings i.e., TCP/IP number, Domain name etc.

Printer:

With this feature you can add new printer software.

System:

This is one of the most important features in the control panel. This contains information about the whole computer. Software, Hardware etc. You can also find if all the hardware devices are fit properly through the Device Manager option.

OBSERVATIONS: -The various features available in the control panel settings has been identified.